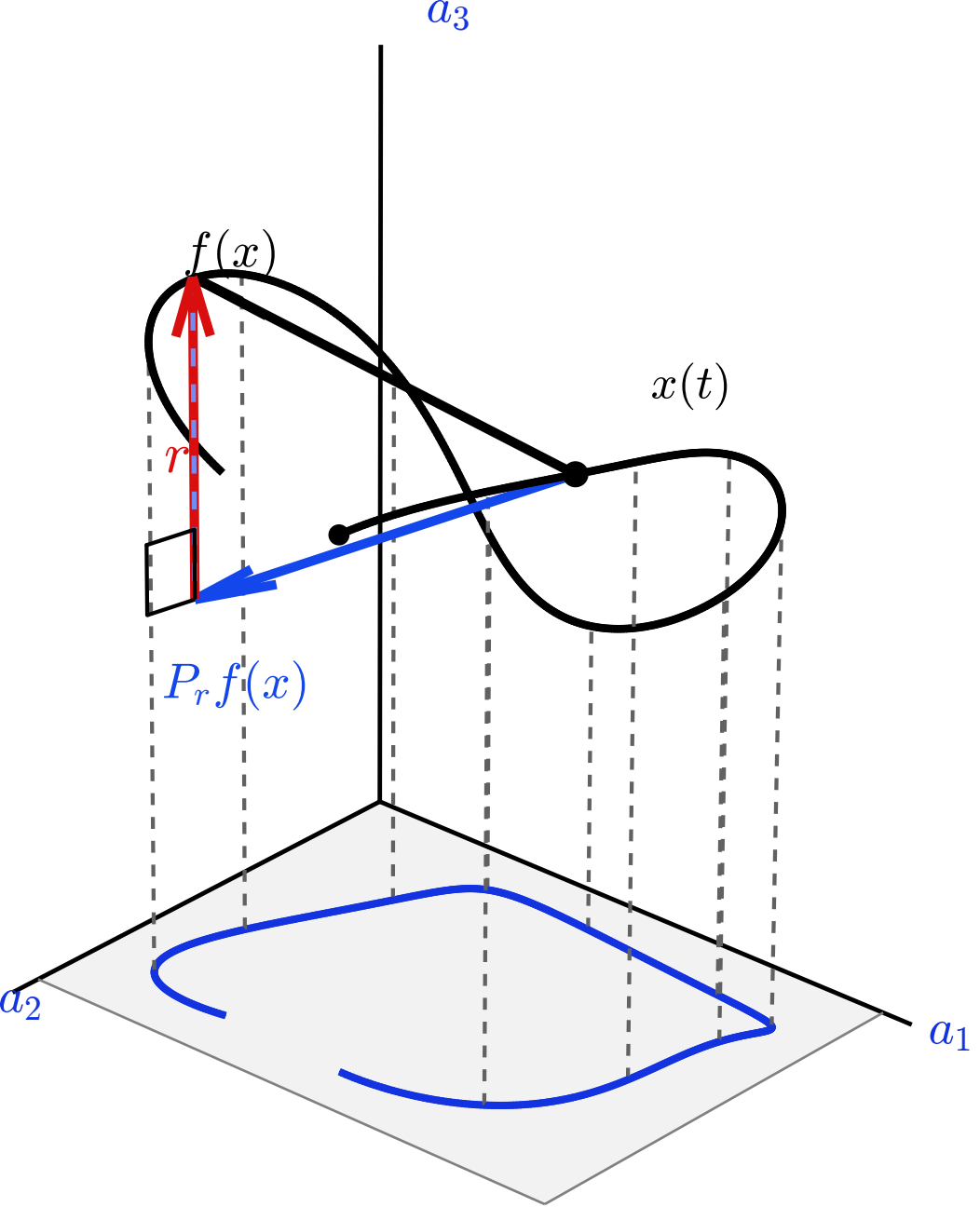

Galerkin condition:

$P_r = U_r U_r^\top$

$r = f(x) - P_r f(x)$

$U_r^\top \, r = 0$

$r \perp \mathcal{U}_r$